INCH-POUND

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MILITARY SPECIFICATION

MASK, FACE, COMBAT VEHICLE CREWMEN'S FLAME, DUST, WIND AND FRAGMENTATION PROTECTIVE

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-M-44128. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers a flame, dust, wind and low-level fragmentation protective face mask consisting of goggles, a detachable lower face piece with bib, and a removable forehead piece. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All government administrative and surveillance procedures applicable to LSC&E items shall be involved in accordance with the contract or purchase order (see 6.2).

1.2 Classification. The face mask shall be of one type and in the following sizes:

Sizes

Small
Medium/Large

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8470

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50195	-	Thread, Aramid
DDD-L-20	-	Label, for Clothing, Equipage, and Tentage (General Use)
PPP-B-26	-	Bag, Plastic, (General Purpose)
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-342	-	Cloth, Wind Resistant, Poplin, Cotton
MIL-F-495	-	Finish, Chemical, Black, for Copper Alloys
MIL-W-5664	-	Webbing, Textile, Elastic, Cotton
MIL-F-21840	-	Fastener Tapes, Hook and Loop, Synthetic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-T-43566	-	Tape, Textile, General Purpose, Natural or in Colors
MIL-W-43685	-	Webbing, Tape, Textile, Aramid Fiber
MIL-G-43914	-	Goggles, Sun, Wind and Dust: 1974
MIL-C-44050	-	Cloth, Ballistic, Aramid
MIL-C-83429	-	Cloth, Plain and Basket Weave, Aramid

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

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MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

THE COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

Department of Defense Standard Color Card for Sewing Threads

(Color cards may be available from the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publication are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

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3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Cloth, ballistic, aramid, water repellent. The cloth for the three ballistic layers shall be water repellent aramid ballistic cloth conforming to type I, class 2 of MIL-C-44050.

3.3.2 Cloth, aromatic polyamide, non-melting. The cloth for the outer layer of the face piece, for the forehead and throat pieces, nose flap, and bib shall be plain weave, aromatic polyamide, non-melting cloth conforming to type II, class 3 of MIL-C-83429 except that the color shall be Olive Green 106.

3.3.3 Cloth, poplin, cotton, flame retardant. The cloth for the innermost layer of the face piece shall be wind resistant cotton poplin, flame retardant treated cloth, Olive Green 107 conforming to class C of MIL-C-342.

3.3.4 Webbing, elastic, cotton. The cotton elastic webbing in the strap shall be 1 inch wide, OG 106, conforming to type I, class 1 of MIL-W-5664.

3.3.5 Tape, textile, cotton. The tape for the non-elastic part of the strap shall be cotton tape, 1 inch wide, Olive Green 106 or Olive Green 7 conforming to type I, class 3 of MIL-T-43566.

3.3.6 Tape, textile, aramid fiber. The tape for tying the air filter to the face piece shall be aramid fiber tape, 1/4 inch wide, OG 106, conforming to type II of MIL-W-43685.

3.3.7 Fastener tape, hook and loop. The hook and loop fastener tapes shall conform to type I, class 2, color green 3421, of MIL-F-21840. The widths shall be as specified in table II.

3.3.8 Thread, aramid. The thread for all stitching shall be aramid thread conforming to A-A-50195, size B, color Olive Drab S-1, C.A. 66022.

3.3.9 Slide. The slide for adjusting the neck strap shall be brass, 0.050 inch thick, conforming to Waterbury Buckle Co. part No. 8090 or equal and shall have a black chemical finish conforming to MIL-F-495.

3.3.10 Goggles, sun, wind and dust. The goggles shall conform to class 3 or 4 of MIL-G-43914.

3.3.11 Labels. The face piece shall have a 1-1/2 by 1-1/2 inch combination size, identification and instruction label conforming to type VI, class 14 of DDD-L-20. The label shall show fastness to accelerated laundering. Format and content of the label shall be as follows:

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SIZE _____.
FACE MASK
COMBAT VEHICLE CREWMEN'S
FLAME, DUST, WIND,
FRAGMENTATION PROTECTIVE

WASHING INSTRUCTIONS

Hand wash in warm water using a mild detergent. Do not wring or twist. Rinse in clear water and hang to dry at room temperature. Do not starch or dry clean. Read your CVC Clothing System Instruction Booklet.

3.4 Design. The face mask shall consist of two major components: (1) goggles having a ballistic protecive lens, hook fastener tape at the lower edge and a removable forehead piece; (2) a lower, soft, close-fitting face piece with its upper edge contoured to the shape of the lower goggle that attaches to the goggle with loop fastener tape. The face piece shall have an adjustable neck strap, a bib, and a cloth air filter. The cloth air filter shall be attached in the neck area with three ties; and when it is not in use, it is rolled, tucked under the chin, and held with hook and loop fastener tape. (see figure 1)

3.5 Patterns. Standard patterns, which provide an allowance of 3/8 ± 1/16 inch unless otherwise specified for all seams will be furnished by the Government and shall be used by the contractor as a guide for cutting working patterns only. The standard patterns shall not be altered in any way and working patterns shall be cut to conform exactly with the standard patterns.

3.5.1 Pattern parts. The pattern components for the face mask shall be cut from the materials as specified and according to the number of parts listed in table I.

TABLE I. List of pattern parts

Item	Nomenclature of part	Cut parts
Cloth, aromatic polyamide	Face piece, outer	1
	Chin, outer	1
	Bib	1
	Throat protector, inner 1/	1
	Throat protector, outer	1
	Nose facing 1/	2
	Air filter	1
	Nose flap, outer	1
	Forehead piece, inner and outer	2
Cloth, ballistic, aramid	Face piece, ballistic filler wide	1
	Face piece, ballistic filler narrow	2
	Chin, ballistic filler wide	2

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TABLE I. List of pattern parts

Item	Nomenclature of part	Cut parts
Cloth, ballistic, aramid (cont'd)	Chin, ballistic filler narrow	1
	Throat protector, ballistic filler wide	1
	Throat protector, ballistic filler narrow	2
	Nose flap, ballistic filler	3
	Forehead piece, ballistic filler	3
Cloth, poplin, cotton	Face piece, inner	1
	Chin, inner	1
	Nose flap, inner 2/	1
Fastener, tape, loop	Forehead piece	1

1/ Cloth, poplin, cotton (inner material) may be used for this piece.

2/ Same pattern part as nose flap inner.

3.6 Construction. The construction shall conform in all respects to the requirements specified in table II and herein. Figure 1 is furnished solely for guidance and information. Should variation from this document appear in figure 1, this document shall govern.

3.6.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. Where stitching crosses seams, each side of the seam shall be spread apart and flat.

3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

3.6.1.1.1 Repairs of type 301 stitching.

a. When thread breaks or bobbin run-outs occur during stitching, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

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b. Except for prestitching, thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Bartacks. Unless otherwise specified, all bartacks shall be 3/8 ± 1/16 inch long and 1/8 ± 1/32 inch wide, and shall contain 28 stitches. Bartacking shall be free from thread breaks and loose stitching.

3.6.1.3 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table II, providing that the seam and stitch type are as specified and the finished component conforms to the required configuration. When a government finished shaper pattern is forwarded, the component shall conform to that pattern.

3.6.1.4 Thread ends. All thread ends that are visible on the finished mask shall be trimmed to a length of not more than 1/4 inch.

3.7 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the face piece and forehead piece.

3.8 Manufacturing operations requirements. The face piece and forehead piece shall be manufactured by and with the use of all operations specified in table II. The contractor is not required to follow the exact sequence of operations. Any additional holding stitching to facilitate manufacture is permissible provided that during assembly the stay stitching is 1/4 ± 1/16 inch from edge of ballistic filler. Unless otherwise specified all fastener tape pieces shall be stitched on all four sides with stitching 1/16 to 1/8 inch from the tape edge. Ends of 1/4 inch wide aramid tape shall be fused.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. The face mask components shall be marked and cut in strict accordance with patterns furnished which show size, cutting directional lines, placement of components and notches for proper assembling of all parts. A cutting tolerance of 1/8 inch is permitted for all parts covered by a pattern. b. The fabric components visible on the exterior of the face mask shall not be drilled for positioning any part except where a component part, when attached, will completely and permanently cover the drill hole. The use of drill holes on the ballistic material is prohibited. c. Each webbing or fastener tape component shall be cut in the required length as specified or as indicated on the patterns ± 1/8 inch unless otherwise specified. d. Cut 1 inch wide loop fastener tape as follows: 1. For the inner face piece (operation 4a), cut six pieces. The four long pieces shall finish 1-3/4 inches on each long side with cut ends 30 degrees from the perpendicular as indicated on the patterns and the two short pieces shall finish 1-1/8 inches long with the ends cut square. 2. For the strap attachment (operation 12f), cut one piece 2 inches long. 3. For the nose flap (operation 7d), cut one piece 1 inch long.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) 4. For the air filter holder (operation 9a), cut one piece 2-1/4 inches long. e. Cut 1-1/2 inch wide loop fastener tape for the outer face piece (operation 5a), as follows: 1. For size M/L, cut two pieces, 3 inches long. 2. For size small, cut two pieces, 2-1/2 inches long. f. Cut 1 inch wide hook fastener tape as follows: 1. For the neck strap (operation 10a), cut one piece 3 inches long. 2. For the air filter (operation 8d), cut one piece 1 inch long. 3. For the air filter, cut two pieces, 3 inches long. The 1/2 ± 1/16 inch wide hook fastener tape shall be from 1 inch wide tape. h. Cut six pieces of 1/4 inch wide aramid tape, 5 inches long, for attaching the air filter (oeprations 6c and 8b). i. Cut one piece of elastic webbing, 7 inches long, for the neck strap (operation 10b). j. Cut cotton webbing for the neck strap (operations 10a and 10d) as follows:						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd)						
	1. One piece 12-1/2 inches long.						
	2. One piece 3 inches long.						
2.	Replacement of defect components. During the spreading, cutting and manufacturing process, components having material defects or damage that are classified as defects in 4.4.2 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Make and assemble ballistic filler.						
	a. With the wide ply of the face piece on the bottom, align the top and sides of the three plies and stitch 3/4 ± 1/4 inch from top and side edges (holding stitch).	301	SSa-1	6-8	B	B	
	b. Overlap the bottom center of the face piece (center chin area) 1/2 ± 1/16 inch and stitch with one row of stitching down the middle of the overlap.	301	LSa-1	8-10	B	B	
	c. Stitch the bottom of the face piece 1/8 inch from the edge of the narrow ply.	301	LSa-1	8-10	B	B	
	d. Center the narrow ply of the chin piece on the wide plies and stitch around the periphery 1/8 inch from the edge of the narrow ply.						
	e. Align the bottom and side edges of the narrow plies of the throat protector on the wide ply and stitch 3/4 ± 1/4 inch from bottom and side edges (holding stitch).	301	SSa-1	6-8	B	B	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Make and assemble ballistic filler. (cont'd)						
	f. Stitch the top of the throat protector 1/8 inch from the edge of the narrow ply.	301	LSa-1	8-10	B	B	
	g. Center and overlap the wide plies of the face and chin (convex side) pieces so that they butt against each others' narrow plies (see figure 1). Stitch with two rows of stitching: one row on each of the overlapped parts 1/8 inch from the edge of the wide plies.	301	LSa-1	8-10	B	B	
	h. Center and overlap the wide plies of the chin (concave side) and throat protector pieces and stitch as in operation g.	301	LSa-1	8-10	B	B	
4.	Make inner mask.						
	a. Stitch 1 inch wide loop fastener tape to the inner face piece as indicated on the pattern.	301	LSbj-1	8-10	B	B	
	b. Stitch the bottom center of the face piece (center chin area).	301	SSa-1	8-10	B	B	
	c. Center and stitch the chin piece (convex side) to the bottom of the face piece with the seam finishing toward the face side of the mask.	301	SSa-1	8-10	B	B	
	d. With the nose hole openings aligned, stitch one nose facing to the face side of the face piece 1/8 inch from the hole opening. Turn facing and smooth out seams.	301	SSa-1	8-10	B	B	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make outer mask.						
	a. Stitch 1-1/2 inch wide loop fastener tape, as indicated on the pattern.	301	LSbj-1	8-10	B	B	
	b. Stitch the bottom center of the face piece (center chin area).	301	SSa-1	8-10	B	B	
	c. Center and stitch the chin piece (convex side) to the bottom of the face piece with the seam finishing toward the outside of the mask.	301	SSa-1	8-10	B	B	
	d. With the nose hole openings aligned, stitch one nose facing to the outside of the face piece 1/8 inch from the hole opening. Turn facing and smooth seams.	301	SSa-1	8-10	B	B	
6.	Make bib.						
	a. With the bottom and sides of the throat protector aligned, stitch around the sides and bottom leaving the top open. Turn, smooth out all seams.	301	SSa-1	8-10	B	B	
	b. Bind the periphery of the bib except the top edge with 1-1/4 inch wide bias binding tape of self material. The binding shall finish 3/8 ± 1/16 inch wide.	301	BSc-1	8-10	B	B	
	c. Align and stitch top edge of the narrow throat protector piece to the top edge of the bib, catching the air filter holder (see operation 9a) and the other two side air filter ties in the seams as indicated on the pattern. Tacked end of holder with the pile piece on top, and one end of ties shall be even with the raw edge of the bib. Holder and ties may be tacked prior to stitching. Holder and ties shall finish to the outside of the mask.	301	SSa-1	8-10	B	B	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make bib. (cont'd)						
	d. Stitch the combination label to the same side of the bib as the throat protector centered from side to side and with the top of the label 1 ± 1/4 inch below the throat protector. Stitching shall be 1/8 inch from edges on all four sides.	301	LSbj-1	8-10	B	B	
7.	Make nose flap.						
	a. Center the three filler plies on the nose flap, innder and stitch around the periphery 1/4 inch from the filler edge (holding stitch).	301	LSbj-1	6-8	B	B	
	b. Stitch two pieces of the 1/2 inch wide hook fastener tape as indicated on the pattern to the nose flap, inner.	301	LSbj-1	8-10	B	B	
	c. With the outer sides of the flap facing each other, stitch around the periphery leaving a 3-1/2 inch long opening at the bottom. Turn and smooth, pushing the filler into all corners. Turn in and top stitch opening and periphery of the flap 1/8 inch from folded edge.	301	SSa-1 & SSe-1	8-10	B	B	
	d. Stitch the 1 inch long piece of loop fastener tape to the side opposite the hook tape side. The loop tape shall be at the center top of the flap with one edge even with the edge of the flap.	301	LSbj-1	8-10	B	B	
8.	Make air filter.						
	a. As indicated on the patterns, make seven darts and stitch down so all darts point toward the center. The dart folds shall be the inside of the air filter.	301	SSa-1	8-10	B	B	

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					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make air filter. (cont'd)						
	b. Tack three of the 5 inch long pieces of 1/4 inch wide aramid tape on the inside bottom of the air filter as indicated on the pattern, with the end of the tapes even with the raw edge of the air filter.	301	SSa-1	6-10	B	B	
	c. Bind the periphery of the air filter with 1-1/4 inch wide bias binding tape of self material, catching the tapes in the stitching. The binding shall finish 3/8 inch wide.	301	BSc-1	8-10	B	B	
	d. On the inside top center of the filter, stitch the 1 inch square piece of hook fastener tape with the corners 1/16 to 1/8 inch from the edge of the filter.	301	LSbj-1	8-10	B	B	
	e. Stitch the two 3 inch long pieces of 1/2 inch wide hook fastener tape to the inside of the air filter with one piece on each side of the center piece. One end of each piece shall be 1-1/2 ± 1/8 inches from the nearest edge of the center piece and the side pieces shall be 1/16 to 1/8 inch from the edge of the filter.	301	LSbj-1	8-10	B	B	
9.	Make air filter holder.						
	a. With one end of a 1/4 inch wide piece of aramid tape centered and sandwiched between one piece each of the 2-1/4 inch long hook and loop fastener tape, stitch the three parts together 1/4 ± 1/16 inch from the end of the assembly. The loop fastener tape shall face the back side of the hook tape and all parts at the stitched end shall be even.	301	SSa-1	6-8	B	B	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Make neck strap.						
	a. Stitch the long edges of the 3 inch long piece of 1 inch wide hook fastener tape to the 4-1/2 inch long piece of cotton non-elastic webbing so that it is 1/2 inch from one end and 1 inch from the other end. Turn the 1 inch end over twice 1/2 inch so that the webbing covers 1/2 inch of the hooks and box stitch 7/16 inch wide.	301	LSbj-1	8-10	B	B	
	b. Place one end of the elastic webbing on the 1/2 inch long extension so that it butts with the hook tape. Turn the cotton webbing 1/2 inch over the elastic and the hooks. Turn the elastic webbing back over end of the cotton webbing so that 1/2 inch of the end is turned under the folded 1/2 inch of cotton webbing and box stitch 7/16 inch wide.	301	EFa-1	8-10	B	B	
	c. Feed 1-1/2 $\pm$ 1/4 inches of the free end of the elastic webbing around the center bar of the metal slide so that the raw end comes out on the side opposite the hooks. Turn the end under 1/2 inch and box stitch 7/16 inch wide to the strap so that the length of the loop to the end of the turn under is 1 $\pm$ 1/4 inch.	301	EFb-2	8-10	B	B	
	d. With the hook side down, feed a few inches of the 12-1/2 inch long cotton webbing, in the direction toward the hook end, through both slots of the slide then turn back through the first slot. Insure that the short adjusting end is opposite the hook side. Turn the fed-through end 1/2 inch twice and box stitch 7/16 inch wide.	301	EFb-2	8-10	B	B	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Make forehead piece.						
	a. Secure the three layers of the filler with a single row of stitching 4 to 5 inches down the middle (holding stitch).	301	SSa-1	6-8	B	B	
	b. Center the filler on one of the cover pieces and stitch together with one row of stitching 1/8 inch from the edge of the filler.	301	SSa-1	8-10	B	B	
	c. With the face sides together, join the cover pieces with a single row of stitching 1/4 inch from the edge of the cover pieces except for about 5 inches at the top.	301	SSa-1	8-10	B	B	
	d. Turn the forehead piece, turn in the raw edges and smooth out any lumps exspecially in the corners. Top stitch opening 1/8 inch from the edge.	301	SSa-1 & SSe-1	8-10	B	B	
	e. Position the loop fastener tape piece on the forehead piece with bottom edges even as shown on the pattern and top stitch 1/8 inch around the edge of the fastener tape piece.	301	LSbj-1	8-10	B	B	
12.	Assemble face mask.						
	a. Center the assembled filler on the back side of the inner so that the filler is between the nose facing and the inner face piece. Stitch through all plies 1/4 ± 1/16 inch around the periphery of the nose opening (holding stitch).	301	SSa-1	6-8	B	B	
	b. Place the outside of the outer face piece on the outside of the inner face piece with all edges aligned and stitch around the sides and top leaving the	301	SSa-1 & SSe-1	8-10	B	B	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Assemble face mask. (cont'd)						
	bottom and 1 inch on each side from the bottom open. Turn and push out all corners and points. Stitch around the periphery of the nose hole 1/8 inch from the edge. Align the inner and outer nose opening edges. Nose opening shall finish flat without twisting or puckering.						
	c. Insert throat protector filler between wide and narrow throat protector, push to edges, smooth and top stitch to the bib 1/4 inch from the bottom and side edges and 1/2 inch from the top row edge of the wide throat protector piece.	301	SSa-1	8-10	B	b	
	d. Smooth the wide throat protector raw edge into the chin piece and turn the raw edge of the inner chin piece and sides 3/8 inch. Stitch through all plies 1/8 from the edge of the turned piece so that the stitching is superimposed on the corresponding seam on the outside of the mask.	301	LSd-1	8-10	B	B	
	e. Smooth filler to the edges and top stitch each side and top of the masks with one row of stitching 1/4 inch from the edge of the mask.	301	SSe-1	8-10	B	B	
	f. Stitch the 2 inch long piece of 1 inch wide loop fastener tape to the outside of the mask with the long edge 1/8 $\pm$ 1/16 inch above the chin/face piece seam and one end 1/8 $\pm$ 1/16 inch from the right edge of the face piece.	301	LSbj-1	8-10	B	B	
	g. Stitch the nose flap to the face piece. With the hook tape facing down, "cup" the nose flap and stitch it to the face piece so that the top center edge of	301	LSd-1	8-10	B	B	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Assemble face mask. (cont'd)						
	the nose flap finishes 1/2 ± 1/16 inch from the top center edge of the face piece with the corners 1/8 ± 1/16 inch from the upper edge of the face piece. Stitching shall be 1/8 inch from edge of flap and extend down 1/2 inch on side of flap, tacking ends.						
	h. Attach the neck strap. Turn the raw end of the strap 1/2 inch and stitch to the outside of the face piece with a 7/16 inch wide box stitch. The long side of the strap shall be 1/8 ± 1/16 inch above the chin/face piece seam with the folded edge 3/4 ± 1/8 inch from the left edge of the face piece. The adjustment end of the cotton strap must face out with the hook side of the attachment end facing in to engage the loop on the face piece without twisting the strap. Note the hook end of the strap with the loop on the face piece.	301	EFa-2	8-10	B	B	
	i. Superimpose the innder and outer chin/face piece seams and tack at the center with a 1/2 ± 1/16 inch bartack.	301	Bartack	18-22	B	B	
13.	Clean face mask.						
	a. Trim and remove all thread ends, and ballistic cloth yarns.						
	b. Remove all spots, stains, and size markings.						
14.	Attach air filter.						
	a. With the hook tape side of the air filter facing up, place the air filter on the outside of the bib and tie each pair of ties with a tight bow knot.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Attach air filter. (cont'd) Fold the air filter in half and roll it tightly between the fastener tapes. Place the forehead piece between the fastener tapes (for storage) and mate the tapes."						

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3.9 Workmanship. The finished face mask shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the specified quality levels. Utmost care shall be taken during component and sub-assembly fabrication to ensure quality workmanship and safety of the service person using the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.2 In-process visual examination of cut parts, ballistic fillers, and patterns. All cut parts for the ballistic filler components shall be examined during the cutting process to determine that parts are cut properly with respect to size and material directional requirements, and textile parts containing defects such as a hole, cut, smash, float, loose slub, hard crease or wrinkles are removed from production. All ballistic fillers shall be examined during assembly of the individual groups to assure that they contain the correct amount and are marked correctly as to the size and the number of plies. In addition to the above, inspection shall be made of the working patterns to ensure that they conform to the Government patterns in all respects. Whenever nonconformance is noted, correction shall be made to the item or items affected and lot in process. Parts which cannot be corrected shall be removed from production.

4.4.3 End item visual examination. The end items shall be examined for defects listed in table III. The lot size shall be expressed in units of masks. The sample unit shall be one mask. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15.0 for total (major and minor combined) defects.

TABLE III. End item visual defects.

Examine	Defect	Classification Major	Minor
Material defects and damages			
Cloth	Any hole, (except drill hole), cut or tear.	101	
	Any exposed drill hole.		201
	Any abrasion marks, broken or missing yarns, or multiple floats.	102	
	Any mend, darn, or patch.	103	
	Any needle chew.	104	
Cloth, binding	Any hole, cut, or tear.	105	
	Not cut on a bias.	106	
Fastener tape	Any hole, cut, or tear.	107	
	Hooks crushed effecting closure.	108	
Webbing and tape	Any hole, cut, tear or frayed edge.	109	
	Needle chew:		
	- Up to 1/8 inch in length.		202
	- More than 1/8 inch in length	110	

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TABLE III. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitchings			
Open seams	Up to and including 1/2 inch.		203
	More than 1/2 inch.	111	
	NOTE: A seam shall be classified as an open seam when one or more stitches joining a seam are broken, or when two or more consecutive skipped or runoff stitches occur.		
Raw edges (except where required or otherwise classified herein)	More than 1/4 inch when securely caught in stitching.		204
	NOTE: Raw edges not securely caught in stitching shall be classified as open seams.		
Seam and stitch type	Wrong seam or stitch type.	112	
	Seam pleated or excessively puckered, clearly noticeable.		205
Stitch tension	Tension loose, resulting in loose bobbin or top thread.		206
	Tension excessively tight, resulting in puckering of material.		207
Stitches per inch (all types of stitching)	One stitch less than minimum specified. 1/		208
	Two or more stitches less than minimum specified. 1/	113	
	Two or more stitches in excess of maximum specified. 1/		209
	NOTE: Variation in the number of stitches per inch caused by operator speeding up the machine and pulling the material in order to sew over heavy places, or in turning corners shall be classified as follows:		

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TABLE III. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitchings (cont'd)			
Stitches per inch (all types of stitching) (cont'd)	Within the minor defect classification - no defect. Within the major defect classification - minor defect.		
1/ Defects to be scored only when the condition exists for 3 inches or more or in several areas with an accumulated distance of 5 inches or more. Applicable to individual seams.			
Rows of stitching	Any row omitted.	114	
Thread breaks, skipped stitches or run-offs (on type 301 stitching)	Overstitched less than 1/2 inch in each direction beyond defective area.		210
	NOTE: On all types of stitching thread breaks or two or more consecutive skipped or run-off stitches not overstitched shall be classified as open seams.		
Stitching ends (on type 301 stitching)	Ends of stitching not secured as specified (except when caught in other stitching or turned under in a hem).		211
Components and assembly	Any component caught in any unrelated operation of stitching (not otherwise classified herein).	115	
	Any component part omitted or not attached as specified (not otherwise classified herein).	116	
General	Any fastener tape reversed, i.e., loop tape attached where hook tape is required or vice versa.		212
	End of strap not properly located or box-stitched.	117	
	Strap attached with hook side of fastener tape facing wrong direction.	118	

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TABLE III. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitchings (cont'd)			
General (cont'd)	End of binding tape not turned under.		213
	Slide bent or broken.	119	
	Slide not proper odor.		214
	Strap not properly threaded through slide.		215
Fit of ballistic fillers	Any bunching.	120	
	Filler smaller than cover by more than 1/4 inch.		216
	Any filler incorrect in size.	121	
Cutting	Any component part not cut in conformance with directional lines on patterns.		217
Cleanness	Any spot or stain clearly noticeable.		218
	Thread ends or ballistic cloth yarns ends not trimmed throughout, as specified.		219
Labels	Omitted, incorrect, illegible or not as specified.		220
Goggles:			
Workmanship and assembly	Frame obviously malformed or otherwise impaired.	122	
	Any component missing.	123	
	Snap fastener does not function properly.	124	
	Ends of elastic webbing not dipped.		221
	Wrong lens in goggles.	125	
	Fastener tape hook loose or crushed.	126	

4.4.4 Packaging inspection. The fully packaged end items shall be examiend for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Inside dimensions of shipping container not as specified.
Content	Number of face masks per shipping container is more or less than required. 1/

1/ For this defect, one shipping container in the sample shall be examined.

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit laod, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A preservation. Each face mask shall be folded to approximately 8-1/2 inches by 4-1/2 inches and together with a forehead piece shall be inserted into a close fitting clear polyethylene bag conforming to type II, style 1 of PPP-B-26. The label on the bib of the face mask shall be visible after insertion. The bagged face mask together with one of the goggles, preserved in accordance with the level A requirements of MIL-G-43914, shall be unit packed in a folding paper-board box conforming to variety 1, style X of PPP-B-566.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Thirty complete face masks, preserved as specified in 5.2.2 and 5.2.3, shall be packed in a nailed wood shipping container conforming to class 2, style 2 or 4 of PPP-B-621. The face masks shall be packed five in length, three in width, and two in depth within a shipping container. Inside dimensions of each shipping container shall approximate 27-1/2 inches in length, 13-1/2 inches in width, and 17 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed and strapped in accordance with PPP-B-621. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Thirty complete face masks, preserved as specified in 5.1, shall be packed in fiberboard shipping containers conforming to style RSC, grade V3c of PPP-B-636. The face masks shall be packed five in length, three in width, and two in depth within a shipping container. Inside dimensions of each shipping container shall approximate 27-1/2 inches in length, 13-1/2 inches in width, and 17 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III and reinforced as specified with the appendix of PPP-B-636.

5.2.3 Commercial packing. Face masks, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), coveralls packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type I or Ia (as specified) of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 90 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking requirements by the contract, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The face mask is intended for use by combat vehicle crewmen for protection against flame, dust, wind and low velocity shell and grenade fragments. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All government administrative and surveillance procedures applicable to LSC&E items shall be invoked in accordance with the contract or purchase order.

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6.2 Acquisition requirements. Acquisition specifications must specify the following:

a. Title, number and date of this specification.
b. Special provisions for verification inspection of Life Support Clothing and Equipment (see 1.1).
c. Size required (see 1.2).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is required (see 3.1.2, 4.3 and 6.4).
f. Level of packing (see 5.2).
g. Type and class of unit load (see 5.2.1).
h. When palletization is required (see 5.3).

6.3 Samples. For access of sample mask, and standard shade samples, address the contracting activity issuing the invitation for bid or request for proposal.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents, regarding arrangements for selection, inspection, and approval of the first article.

6.5 Subject term (key word) listing.

Goggles
Life support
[illegible]

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the last previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Preparing activity:

Army - GL

(Project 8470-A145)

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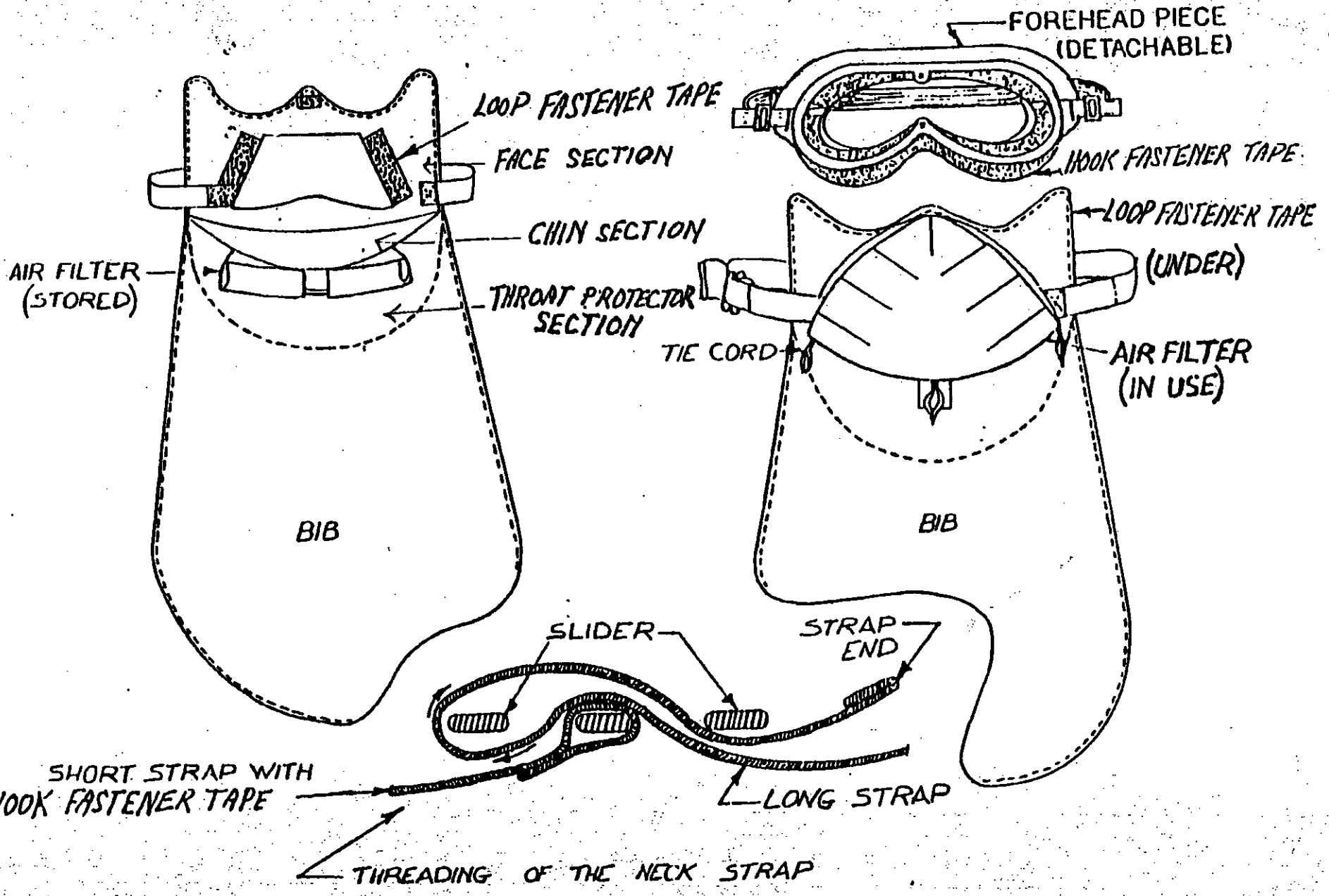


FIGURE 1. MASK, FACE, COMBAT VEHICLE CREWMEN'S

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